SkinSeal
just seal & heal

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Skinseal™

(N-Butyl-2-Cyanoacrylate)





"Delivering Confidence, Sealing Trust."

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ADVANCED WOUND CLOSURE

SkinSeal : N-Butyl-2 Cyanoacrylate Topical Skin Adhesive

Comprehensive Clinical Overview: Why Choose Skin Seal?

Skinseal is a state-of-the-art synthetic, biocompatible tissue adhesive. Composed of N-Butyl-2 Cyanoacrylate (NBCA) in a ready-to-use 0.25 ml configuration, it offers a superior alternative to traditional sutures and staples for dermal closure, Aligning with the growing emphasis on maximizing operative efficiency, Improving cosmesis, and enhancing patient recovery.

NBCA is a fast-polymerizing, Monomeric adhesive that reacts with anionic substances (like moisture or blood) to form a strong, flexible, and waterproof polymer film that effectively approximates wound edges. This unique mechanism of action is supported by extensive clinical evidence demonstrating advantages across major surgical fields, including general surgery, obstetrics, and emergency.

Evidence-Based Advantages of N-Butyl-2 Cyanoacrylate (Skinseal)

Clinical Benefit	Key Findings & Supporting Studies
I. Operative Efficiency	**Significantly Reduced Closure Time:** NBCA closure time is substantially shorter than conventional suturing, Stream lining operative work flow. Closure time averages were reduced by 30–40% (4.73 min vs. 7.10 min for sutures, $p<0.001$) *
II. Patient Comfort & Pain	**Reduced Postoperative Pain:** Patients experience significantly lower pain scores due to the absence of suture-related skin tension and decreased mechanical irritation.
III. Wound Healing & Infection	**Microbial Barrier and Lower Infection Rates:** NBCA exhibits bacteriostatic properties and forms a waterproof seal, acting as a physical barrier against microbial invasion and reducing bacterial penetration. Infection rates were significantly lower in NBCA groups. Histological studies show it causes a less intense inflammatory reaction than sutures.
IV. Cosmetic Outcome	**Superior Aesthetics:** NBCA demonstrates superior short- term cosmetic out comes by laminating suture track marks. Long-term cosmetic results (evaluated at 3 months or more) are comparable to high-quality subcuticular sutures.
V. Convenience	**No Suture Removal Required:** The adhesive spontaneously sloughs off, eliminating the need for follow-up visits specifically for removal and improving patient satisfaction and compliance. Studies demonstrate NBCA is more effective overall due to reduced operative time and lower postoperative care requirements.

* A Comparative Study of N-Butyl Cyanoacrylate with Suture Material for Skin Closure in Open Inguinal Hernia Repair in a Tertiary Care Centre.

Applicable Surgical Areas

Skinseal (N-Butyl-2-Cyanoacrylate) is indicated for topical skin closure in procedures where the wound edges are well-approximated and tension is low to moderate.

◆ General Surgery ◆ Gynecology ◆ Pediatric Laceration ◆ Emergency First Aid
◆ General & Minimally Invasive ◆ Minor Cuts And Abrasions

Technical Application Notes (0.25ml Ampoule)

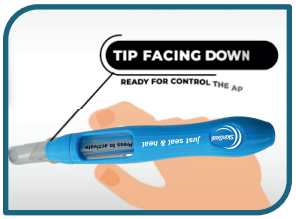
Wound Preparation : The wound site must be clean and dry to ensure optimal bonding and healing. Avoid application over wet areas or bleeding sites, as this accelerates polymerization and increases dehiscence risk. Hemostasis must be achieved prior to application.

Approximation: Subcutaneous layers should be closed (e.g., 80% of maxillofacial patients received subcutaneous sutures), and the skin edges should be manually approximated before application.

Application: Apply the NBCA adhesive in a thin, uniform layer by sliding the neck of the ampule over the wound edge. Excessive application can create a dense, Fragile layer that may peel easily and cause dehiscence.

Polymerization: NBCA polymerizes rapidly upon contact with tissue. Maintain light manual pressure for approximately 30 to 60 seconds to achieve full strength.







Important Shelf Life, Safety & Storage Information

◆ Shelf life : 24 months ◆ For external use only ◆ Keep out of reach of children
◆ If any adverse reaction occurs, Seek medical advice immediately

FAQs-Skin Seal (NBCA)

1. Is a dressing required after Skinseal application?

◆ Use of dressings varies. NBCA forms a protective film; studies report no dressing (e.g., cesarean sections) or short-term dressings (24 hours) depending on procedure.

2. Can Skinseal be applied on wet or bleeding wounds?

◆ No. The wound should be cleaned and dried. Moisture or active bleeding accelerates polymerization and increases dehiscence risk. Approved for fresh, smooth skin wounds only.

3. Can Skinseal replace sutures?

◆ Yes. NBCA provides adequate tensile strength. While sutures show higher early resistance, long-chain cyanoacrylates show superior strength in later healing stages. Firm closure lasts up to 8 days.

4. What if bleeding occurs after application?

- NBCA has strong hemostatic properties. Persistent bleeding indicates inadequate hemostasis and is a contraindication.

5. What if the adhesive sticks unintentionally?

- Cyanoacrylates adhere to most materials. Avoid contact with gloves or instruments. Accidental adhesion can be removed using acetone.

6. Is Skinseal application painful?

- Pain is reduced due to absence of suture tension. Degradation products may provide mild local anesthetic effect.

Post-Operative Care & Healing

7. Can patients shower after Skinseal closure?

- Yes. NBCA forms a waterproof seal. Showering is permitted from the next day in reported studies.

8. Does the adhesive need removal?

- No. NBCA is biodegradable and naturally sloughs off in 7–10 days, Eliminating suture removal visits.

9. Can patients peel off the adhesive?

- No. Premature removal may cause wound dehiscence.

10. What if inflammation or infection occurs?

- NBCA causes less inflammation than sutures and has bacteriostatic properties. Persistent pruritus or erythema may indicate hypersensitivity-consult a doctor.

11. Is there an allergy or hypersensitivity risk?

- Yes. Patients allergic to cyanoacrylates or formaldehyde are contraindicated. HSRs (5.6–13.3%) may occur, Especially in patients sensitive to adhesives.

Product & Professional Use

12. Who can use Skinseal?

- For trained medical personnel only. Contraindications include allergy, Active bleeding, Wound tension, Uncontrolled diabetes, Systemic infection or keloid tendency.

13. Stability after opening the ampoule?

- Single-use only. Store unopened ampoule at 40°C. Discard after usage.

14. Can leftover solution be reused?

- No. One-time use only.

15. Is re-sterilization required?

- No. Skin Seal is Ethylene Oxide sterilized and does not require re-sterilization.

16. Are topical drugs or antibiotics needed post-closure?

- Generally, no routine topical medications or antibiotics are required.

Clinical Evidence & Economics

17. How long has NBCA been used clinically?

- Cyanoacrylates have been used in surgery for over 30 years.

18. What cosmetic results can be expected?

- Studies show better cosmetic outcomes due to absence of suture marks.

19. Is Skin Seal cost-effective?

- Yes. Despite higher unit cost, overall treatment cost is lower due to reduced operative time and no suture removal.

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